BIO Branch Book

Replication, Metabolism, and Certified Biological Structure

Elements of Nested Fibrational Cosmology

Prospective Derived Branch

Contents

1. Purpose and Canonical Seed

This branch asks whether biological structure — self-replication, metabolism, heredity, cellular boundary formation, and Darwinian evolution — can be realized as a lawful descendant of the NFC source architecture. The aim is not to import the standard language of cells, genes, fitness landscapes, proteins, or metabolic pathways as primitives, but to determine when such language is licensed as an effective encoding of certified discrete observable structure within a declared branch regime. This posture is mandated by Book I's no-smuggling stance, Book V's branch-legitimacy law, Book VI's continuum-interface discipline, and Book VII's scoped-certificate law.

The central question is: starting from nothing but a finite class of physical configurations, a finite admissible probe discipline, and the observational equivalence imposed by that discipline, can one derive certified self-replicating structure, metabolic bookkeeping, heritable transport, and the emergence of a boundary-enclosed arena? No cell, gene, fitness landscape, vitalistic force, or biological essence is assumed. Each such notion, if it appears at all, must emerge as a quotient-visible, source-descended object within the declared regime.

Three structural observations motivate the branch within the existing canonical corpus. First, Book II already establishes a *boundary capacity law* — information capacity at interfaces is logarithmically bounded. The biology branch will ask whether a certified observable configuration can develop a *self-organized boundary* that exploits this capacity structure without violating it. Second, Book III already establishes a *unified coercive-transport inequality* governing how stability costs distribute. The biology branch will ask whether a configuration can minimize that transport cost by self-replicating — reproducing its own certified observable class rather than diffusing into equivalence-class noise. Third, the NS branch already treats a *continuation criterion* whose failure marks the singular endpoint; the biology branch treats the analogous quantity — a *replication-continuation criterion* whose maintenance marks the persistence of a living configuration.

The initial endpoint is deliberately modest: certified replication structure, metabolic bookkeeping, and hereditary transport within a declared observable regime. Multicellular organization, evolutionary dynamics, and molecular-biological specificity are deferred as named open obligations.

Remark 1.1 ([R]). **Strategic path.** The shortest canonical route is replication-founded heredity: begin from certified admissible-copy events (the replication kernel), build a metabolic ledger for resource-transformation bookkeeping, show that replication is fidelity-bounded by a positive margin, prove a no-hidden-channel theorem for the hereditary transport map, then ask whether the assembled structure certifiably descends from **U** via the Book V branch-legitimacy law. Evolutionary dynamics and multicellular organization remain downstream open obligations.

2. Imported Spine Structure

Imported spine results. All certified results of Books I–VII are required for lawful descent, transport accounting, continuum-interface discipline, and closure governance. In particular:

(1) *Book I*: observational quotient discipline; no primitive cells, genes, organisms, species, or molecular-biological vocabulary may be assumed. Biological content must be quotient-visible and admissibly testable, not imported as background life-science.
(2) *Book II*: stabilized local outcomes, collar-visible control, and boundary capacity structure. The logarithmic boundary capacity law is the canonical constraint on any self-organized interface claiming to separate an interior from an exterior without hidden information import.
(3) *Book III*: persistence and transport law; additive bookkeeping. The unified coercive-transport inequality governs how stability costs distribute across replication, metabolic transformation, and boundary maintenance — the three canonical cost categories of the biology branch.
(4) *Book IV*: the universal source object **U** and source-descended comparison architecture. All certified biological objects must factor through **U**.
(5) *Book V*: lawful branchhood and branch visibility; the constitutional law under which a biology branch is admitted.
(6) *Book VI*: continuum language licensed only as faithful shorthand for certified discrete structure within a declared interface regime; the main gate for continuous-time population dynamics, thermodynamic quantities, information-rate quantities, and chemical-kinetics notation in formal biological models.
(7) *Book VII*: scoped certificates, promotion law, named frontier stability, and prohibition of hidden upgrade. Claims about universal features of life, infinite evolutionary potential, or species-level universals are subject to this governance without exception.

Imported branch precedents.

(1) *Book II boundary capacity*: the interface constraint that bounds the information capacity of any self-organized boundary; the *membrane template* for this branch.
(2) *NS branch*: the continuation-criterion / obstruction-decay pattern as the backbone of a main frontier theorem; the *replication-continuation template* — a biological configuration either continues to replicate or encounters a frontier analogous to NS blowup.
(3) *SPEC branch*: probe-response objects and transition ledgers; the *metabolic-probe template* for certified resource-transformation bookkeeping.
(4) *LING branch*: the no-smuggling posture on inherited structure; in particular, the principle that a derived inventory (contrast classes, certified copy classes) must emerge from quotient structure, not be imported as a named alphabet or genetic code.

3. Branch-Specific Arena

3.1. Standing Rules.

Standing Rule 3.1 ([D]). (No Smuggling.) No biological object — cell, gene, organism, chromosome, protein, metabolic pathway, fitness value, or species — may be treated as primitive branch data merely because it is familiar in external biological theory. Every branch-visible biological object must be sourced through **U**, the declared descent machinery, and the biology observable family. This is a branch-local instance of Book I no-smuggling and Book V branch-legitimacy discipline.

Standing Rule 3.2 ([D]). (No Vitalism.) No vitalistic force, teleological drive, biological essence, or special-purpose self-organization principle may be assumed. Every instance of self-organized structure that appears in this branch must be derived from quotient-visible observational data and the spine's transport and boundary machinery.

Standing Rule 3.3 ([D]). (Quotient Visibility.) A biological distinction between two configurations counts only when it is quotient-visible under admissible probe tests. In particular, a distinction between a living and a non-living configuration counts only when it is witnessable by some admissible probe within the declared regime — not by appeal to an external biological authority.

Standing Rule 3.4 ([D]). (No Hidden Genetic Code.) No hidden genetic code, underlying molecular representation, or abstract hereditary level may be assumed. Hereditary structure may appear only after a branch-local derivation certifies that such structure faithfully encodes quotient-visible replication data within the declared regime.

Standing Rule 3.5 ([D]). (No Self-Promotion.) A finite replication certificate may not self-promote to a claim about unlimited evolutionary potential, infinite hereditary lineages, or universal features of life. Such promotion requires an explicit transfer theorem. This is a direct Book VII governance consequence.

Standing Rule 3.6 ([D]). (Frontier Visibility.) Named frontiers in replication fidelity, metabolic closure, hereditary transport, boundary self-organization, evolutionary dynamics, or multicellular coordination remain visible until discharged by theorem. Rephrasing does not discharge them.

Standing Rule 3.7 ([D]). (No Fitness Primitive.) No fitness landscape, adaptive value, or selection coefficient may be treated as a primitive. Fitness, to the extent it appears at all, must emerge as a quotient-visible differential replication rate licensed by certified replication and hereditary transport structure.

Standing Rule 3.8 ([D]). (Continuum Gate.) Continuous-time population dynamics, thermodynamic entropy quantities, information-theoretic rates, and chemical-kinetics differential equations may appear only after a branch-local Book VI bridge theorem certifies that they faithfully encode certified discrete biological dynamics within a declared continuum interface regime.

3.2. Declared Target Regime.

DEFINITION 3.9 ([D]). (Biology Target Regime.) A *biology target regime* is a declared branch regime $\mathcal{R}_{\mathrm{Bio}}$ consisting of:

(1) a certified observable class $\mathcal{C}_{\mathrm{obs}}$ of physical configurations — the objects on which replication, metabolic, and hereditary probe tests are performed;
(2) a finite admissible probe discipline $\mathcal{P}$, each element of which is a declared admissible process detecting a quotient-visible distinction among configurations;
(3) a declared resource-transformation window $\mathcal{W}_{\mathrm{met}}$ specifying the range of metabolic operations within which resource-transformation claims have force; and
(4) a declared hereditary comparison discipline governing when two descendant configurations are equivalent for the purpose of hereditary claims.

The role of $\mathcal{R}_{\mathrm{Bio}}$ is to state explicitly where biological claims are supposed to have force.

REMARK 3.10 ([R]). A *physical configuration* in this branch is not yet a cell, bacterium, or organism. At this stage it is only a certified observable arrangement of matter-like objects on which admissible probe tests can be performed. Whether any such configuration qualifies as a replicating entity is determined by the replication kernel, not by prior biological classification.

3.3. Biology Observable Family.

DEFINITION 3.11 ([D]). (Biology Observable Family.) A *biology observable family* on $\mathcal{R}_{\mathrm{Bio}}$ is a triple

$$\mathcal{O}_{\mathrm{Bio}} = \big(\Phi_{\mathrm{Bio}},\, \mathcal{M}_{\mathrm{Bio}},\, \mathcal{H}_{\mathrm{Bio}}\big)$$

where:

(1) Φ_{Bio} is the *replication kernel*: the quotient-visible self-copy data induced on $\mathcal{C}_{\mathrm{obs}}$ by admissible replication-probe tests; it records which configurations produce certified copies of their own equivalence class;
(2) $\mathcal{M}_{\mathrm{Bio}}$ is the *metabolic ledger*: certified records of resource-transformation operations — the admissible processes by which a configuration acquires, converts, and dissipates resources within the declared metabolic window, with full transport bookkeeping; and
(3) $\mathcal{H}_{\mathrm{Bio}}$ is the *heredity-descent object*: a branch-visible certified transport map governing how observable properties are transmitted from a configuration to its certified replication products across declared descent steps.

DEFINITION 3.12 ([D]). (Replication Kernel.) The *replication kernel* Φ_{Bio} is the collection of ordered pairs (X, X') of configurations together with the certified set of admissible replication-probe tests witnessing that X' is a copy of X within the declared regime. Two configurations X and X' satisfy the *replication relation* $X \xrightarrow{\mathrm{rep}} X'$ if and only if some admissible probe certifies that X' shares the quotient-visible equivalence class of X and was produced by an admissible replication process.

REMARK 3.13 ([R]). The replication relation is not defined by molecular mechanism, DNA sequence, or template-copying chemistry. It is defined by admissible probe certification that the product configuration falls in the same observable equivalence class as the source configuration. If, under a given regime, molecular copying turns out to be the minimal admissible witnessing process, that is a derived theorem, not a branch assumption.

Definition 3.14 ([D]). (Replication Fidelity Margin.) The *replication fidelity margin*

$$F_{\mathrm{Bio}}(X)$$

is a branch-specific nonnegative quantity measuring the observable separation between the equivalence class of a configuration X and the equivalence classes of its certified replication products. A positive fidelity margin means the copies are observationally indistinguishable from the parent within the declared regime. This is the biological analogue of the spectral margin of the SPEC branch and the contrast margin of the LING branch.

Remark 3.15 ([R]). The replication fidelity margin is the biological instance of the canonical gap structure: YM mass gap (spectral separation), SPEC spectral margin (resonance isolation), LING contrast margin (phonemic separation), BIO fidelity margin (copy fidelity). In each case the structural claim is a positive separation between inequivalent observable classes on the declared regime.

Definition 3.16 ([D]). (Metabolic Ledger.) The *metabolic ledger* $\mathcal{M}_{\mathrm{Bio}}$ is a certified bookkeeping object assigning to each admissible resource-transformation operation a record of:

(1) the resource equivalence classes consumed (input);
(2) the resource equivalence classes produced (output); and
(3) the transport cost and defect contribution incurred, governed by the Book III unified coercive-transport inequality.

The metabolic ledger is additive over declared transformation steps.

Remark 3.17 ([R]). The metabolic ledger does not assume thermodynamic equilibrium, free-energy minimization, or any imported chemical-kinetics framework. Whether such frameworks are lawfully licensed here is a downstream Book VI bridge question. At this stage the metabolic ledger is a purely discrete accounting object.

Definition 3.18 ([D]). (Heredity-Descent Object.) A *heredity-descent object* $\mathcal{H}_{\mathrm{Bio}}$ is a branch-visible certified transport map that:

(1) is sourced through $\mathbf{U}$;
(2) assigns to each certified replication step $X \xrightarrow{\text{rep}} X'$ a declared observable-class transmission record; and
(3) produces only quotient-visible distinctions among descendant configurations, with all deviations from perfect fidelity accounted for within the declared replication fidelity margin.

$\mathcal{H}_{\mathrm{Bio}}$ is not a genetic code, a chromosome map, or any other pre-given hereditary formalism. It is a derived certified transport object, admitted only when the branch-legitimacy conditions of Book V are satisfied.

DEFINITION 3.19 ([D]). (Biological Boundary Object.) A *biological boundary object*

$$\mathcal{B}_{\mathrm{Bio}}(X)$$

is a branch-visible certified interface structure associated with a configuration X, satisfying the Book II boundary capacity law: information capacity at the interface is logarithmically bounded by the declared interface scale. $\mathcal{B}_{\mathrm{Bio}}(X)$ certifies the observable separation between an interior region and an exterior region for configuration X without importing any background geometry.

REMARK 3.20 ([R]). The biological boundary object is the canonical NFC surrogate for a cell membrane. It is not assumed to be a lipid bilayer or any other chemically specified structure. The only certified property is that it satisfies the Book II boundary capacity law and produces a quotient-visible interior/exterior distinction on the declared regime.

3.4. Endpoint Datum.

DEFINITION 3.21 ([D]). (Biology Endpoint Datum.) The *biology endpoint datum* is the branch-visible package

$$E_{\mathrm{Bio}} = \big(\mathcal{C}_{\mathrm{obs}},\ \mathcal{O}_{\mathrm{Bio}},\ F_{\mathrm{Bio}},\ \mathcal{B}_{\mathrm{Bio}},\ \mathcal{W}_{\mathrm{met}}\big)$$

consisting of the certified observable class, the biology observable family, the replication fidelity margin, the biological boundary object, and the declared metabolic window.

DEFINITION 3.22 ([D]). (Lawful Biology Descendant Claim.) A *lawful biology descendant claim* is a claim that the endpoint datum E_{Bio} is sourced through $\mathbf{U}$ and supports a certified replication, metabolism, heredity, or boundary statement on the declared regime.

3.5. Current Branch Posture.

PROPOSITION 3.23 ([C]). (Biology Branch Candidate.) [Conditional on the biology target regime, observable family, and endpoint datum all being sourced through $\mathbf{U}$, branch-visible, and admitting a Book V legitimacy witness.] *The biology program is a lawful branch candidate.*

PROOF. Immediate from the branch-legitimacy doctrine (Book V) applied to the declared biology endpoint datum. □ □

REMARK 3.24 ([R]). Prop. 3.23 constitutes the branch architecturally. It does not yet certify any replication theorem, metabolic theorem, hereditary transport theorem, or boundary self-organization theorem.

4. Admissible Probes and Replication Signatures

4.1. Admissible Replication Probes.

DEFINITION 4.1 ([D]). (Admissible Replication Probe.) An *admissible replication probe* is a declared admissible process $P \in \mathcal{P}$ whose effect on the certified observable class is branch-visible and accountable within the spine's admissible-test discipline.

Paradigm cases include copy-verification tests, boundary-integrity tests, and resource-consumption tests — but none of these is assumed as a primitive; each must be declared and certified within the NFC framework.

Standing Rule 4.2 ([D]). (No Hidden Probe.) No admissible replication probe may introduce hidden molecular labels, undeclared biochemical representations, or non-certified side channels. Any probe violating this rule is not admissible for biology branch claims.

Definition 4.3 ([D]). (Replication Episode.) A *replication episode* beginning from configuration X is a certified finite sequence of admissible processes — resource acquisition, transformation, copy-assembly, boundary-formation — producing a configuration X', together with the declared transport cost record inherited from the metabolic ledger.

Remark 4.4 ([R]). A replication episode is the biology analogue of a probe episode (SPEC branch) or contrast probe episode (LING branch). It is not a description of a biochemical reaction sequence; it is a certified admissible process record within the NFC framework.

4.2. Replication Signatures and Equivalence.

Definition 4.5 ([D]). (Replication Signature.) A *replication signature*

$$\mathsf{Rep}(X; t)$$

is a branch-visible assignment from a configuration X and declared replication step t to a quotient-visible copy record, together with its transport and fidelity-defect bookkeeping.

Standing Rule 4.6 ([D]). (Signature Invariance.) A replication signature has theorem-bearing force only to the extent that it is invariant under presentation-level differences already eliminated by the observational quotient. Two configurations that are observationally equivalent cannot bear different replication signatures.

Definition 4.7 ([D]). (Replication-Signature Equivalence.) Two replication signatures are *replication-equivalent* if they determine the same quotient-visible copy record under the declared hereditary comparison discipline of $\mathcal{R}_{\mathrm{Bio}}$.

4.3. One-Step Hereditary Ledger.

Definition 4.8 ([D]). (One-Step Hereditary Ledger Entry.) Given a certified replication episode from X to X', the *one-step hereditary ledger entry*

$$\Delta_{\mathrm{her}}\big(X \xrightarrow{\mathrm{rep}} X'\big)$$

records the certified observable-class transmission from X to X', together with any declared fidelity deviation, transport cost, or visible defect contribution inherited from the metabolic ledger.

Remark 4.9 ([R]). The one-step hereditary ledger entry is the biology analogue of the one-step compositional ledger entry (LING branch) and the one-step spectroscopy transition ledger (SPEC branch). In each case the core structure is: take a certified operation, record its effect on the observable equivalence class, account for the transport cost.

5. Core Theorem Shells and Named Frontiers

The following theorem shells articulate the main claims of the biology branch. Each carries the status it would bear *if* its stated open obligations were discharged. The open obligations are named explicitly as branch frontiers.

5.1. Replication Certification.

Open Obligation 5.1 ([C]). (O-BIO.REP: Replication Certification.) Show that the replication kernel Φ_{Bio} is sourced through $\mathbf{U}$ and constitutes a lawful branch observable within the declared regime, with all probe declarations inherited from the Book I admissible-test discipline.

Theorem 5.2 ([C]). (Replication Certification Theorem.) [Conditional on O-BIO.REP: Replication Certification.] *The replication kernel Φ_{Bio} is a lawful branch observable descended from $\mathbf{U}$. The certified replication relation $X \xrightarrow{\text{rep}} X'$ is quotient-visible and admissibly testable on the declared regime.*

Proof. By the branch-legitimacy doctrine (Book V) applied to the declared replication kernel, together with the Book I observational quotient discipline. Full proof is conditional on O-BIO.REP. □ □

5.2. Metabolic Ledger Theorem.

Open Obligation 5.3 ([C]). (O-BIO.MET: Metabolic Bookkeeping.) Show that the metabolic ledger $\mathcal{M}_{\text{Bio}}$ is additive, sourced through the Book III transport accounting, and that all resource-transformation operations preserve transport bookkeeping without hidden defect.

Theorem 5.4 ([C]). (Metabolic Ledger Theorem.) [Conditional on O-BIO.MET: Metabolic Bookkeeping.] *The metabolic ledger $\mathcal{M}_{\text{Bio}}$ is additive over declared resource-transformation steps: transport costs distribute across components without hidden channel.*

Proof. By the Book III unified coercive-transport inequality applied to the resource-transformation operations, together with the no-hidden-probe standing rule (Sr. 4.2). Full proof is conditional on O-BIO.MET. □ □

5.3. Replication Fidelity Margin Theorem.

Open Obligation 5.5 ([C]). (O-BIO.FID: Replication Fidelity Margin.) Show that the replication fidelity margin F_{Bio} is branch-computably positive for certified replicating configurations within the declared regime, and that no fidelity collapse occurs without a declared admissible error event.

Theorem 5.6 ([C]). (Replication Fidelity Margin Theorem.) [Conditional on O-BIO.FID: Replication Fidelity Margin.] *For any certified replicating configuration X, $F_{\mathrm{Bio}}(X) > 0$: the replication products of X are observationally indistinguishable from X within the declared fidelity tolerance. The certified replication inventory is discretely separated from non-replicating configurations on the declared regime.*

Proof. By the observational quotient discipline (Book I) and the declared admissible-test finiteness condition on $\mathcal{R}_{\mathrm{Bio}}$. Positivity of the fidelity margin reflects the separating power of the admissible replication probe discipline. Full proof is conditional on O-BIO.FID. □ □

5.4. No-Hidden-Channel Theorem.

Open Obligation 5.7 ([C]). (O-BIO.NC: no-hidden Hereditary Channel.) Show that no hereditary transport pathway exists outside the declared one-step hereditary ledger: any discrepancy between two descendant configurations sharing the same certified parent would require an undeclared hereditary channel, ruled out by the exhaustive source-image condition.

Theorem 5.8 ([C]). (No-Hidden-Hereditary-Channel Theorem.) [Conditional on O-BIO.NC: no-hidden Hereditary Channel.] *Two descendant configurations sharing the same certified parent and the same admissible replication process have the same replication signature, up to the declared fidelity tolerance. No undeclared hereditary pathway can produce a branch-visible distinction invisible to the declared hereditary ledger.*

Proof. Any discrepancy would require an undeclared dissipative or combinatorial hereditary channel. By the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$ and the no-hidden-gene standing rule (Sr. 3.4), no such channel is admitted. Full proof is conditional on O-BIO.NC. □ □

5.5. Boundary Self-Organization Theorem.

Open Obligation 5.9 ([C]). (O-BIO.BND: Boundary Self-Organization.) Show that the biological boundary object $\mathcal{B}_{\mathrm{Bio}}(X)$ can emerge from the replication kernel and metabolic ledger without importing a background spatial geometry: that a certified configuration can develop a quotient-visible interior/exterior separation satisfying the Book II boundary capacity law, as a derived consequence of replication and metabolic structure.

Theorem 5.10 ([C]). (Boundary Self-Organization Theorem.) [Conditional on O-BIO.BND: Boundary Self-Organization.] *For a certified replicating configuration X with positive fidelity margin and additive metabolic ledger, a biological boundary object $\mathcal{B}_{\mathrm{Bio}}(X)$ exists satisfying the Book II boundary capacity law. The interior/exterior separation is quotient-visible and sourced through $\mathbf{U}$.*

Proof. By the Book II boundary capacity law applied to the declared interface scale of X, together with the replication certification (Thm. 5.2) and metabolic ledger (Thm. 5.4). Full proof is conditional on O-BIO.BND. □ □

Remark 5.11 ([R]). The boundary self-organization theorem is the canonical NFC surrogate for the emergence of a cell membrane. It is not a theorem about lipid bilayers; it is a theorem about the existence of a certified interface structure satisfying the Book II capacity law. Whether lipid bilayers are the minimal physical realization of such an interface in the target regime is a downstream question, not a branch assumption.

5.6. Hereditary Descent Theorem.

Open Obligation 5.12 ([C]). (O-BIO.HER: Hereditary Descent.) Show that the heredity-descent object $\mathcal{H}_{\mathrm{Bio}}$ is sourced through $\mathbf{U}$, passes the five-condition Book V branch-legitimacy test, and transmits observable properties across certified replication steps without hidden channel and within the declared fidelity margin.

Theorem 5.13 ([C]). (Hereditary Descent Theorem.) [Conditional on O-BIO.HER: Hereditary Descent.] *The heredity-descent object $\mathcal{H}_{\mathrm{Bio}}$ is a lawful branch-visible certified transport map. Observable properties are transmitted from X to X' across certified replication steps within the declared fidelity margin, with no hidden hereditary channel.*

Proof. By the Book V branch-legitimacy doctrine applied to $\mathcal{H}_{\mathrm{Bio}}$, together with the no-hidden-gene and no-smuggling standing rules. Full proof is conditional on O-BIO.HER. □ □

5.7. Evolutionary Dynamics.

Open Obligation 5.14 ([C]). (O-BIO.EVO: Evolutionary Dynamics.) Show that differential replication rates across equivalence classes of replicating configurations — a derived quotient-visible quantity — give rise to certified selection dynamics within the declared regime. Fitness is to emerge as a replication-rate differential, not as an imported adaptive value. This obligation is expected to require an iterated hereditary ledger and a Book VI bridge theorem for continuous-time population dynamics.

Remark 5.15 ([R]). **Posture on evolution.** Darwinian evolution in the ordinary biological sense refers to heritable variation, differential fitness, and cumulative change in population composition over many generations. Within the NFC framework, fitness cannot be a primitive; it must emerge as a quotient-visible differential replication rate. Selection dynamics at the population level, if they appear at all, must be licensed by a Book VI bridge theorem as the asymptotic limit of certified multi-step hereditary transport. Whether such a bridge theorem can be proved within the NFC framework is a genuine open question.

5.8. Multicellular Organization.

Open Obligation 5.16 ([C]). (O-BIO.MULTI: Multicellular Organization.) Show, or certify the impossibility of showing within the NFC framework at the current level, that certified boundary-enclosed replicating configurations can coordinate into hierarchically organized multicellular structures — that is, that $\mathcal{B}_{\mathrm{Bio}}$ and $\mathcal{H}_{\mathrm{Bio}}$ can give rise to certified second-order boundary and hereditary objects. This obligation is deeply downstream of O-BIO.BND and O-BIO.HER and is not anticipated at the current architectural stage.

REMARK 5.17 ([R]). **Posture on multicellularity.** Multicellular organization is the deepest downstream obligation in this branch. No claim of multicellular structure is made here. Multicellularity, if it appears at all, must emerge from iterated application of the boundary self-organization and hereditary descent machinery at a higher organizational level. Whether such iteration is lawfully licensed within the NFC framework is a named open question, not a currently certified result.

5.9. Biology Endpoint.

OPEN OBLIGATION 5.18 ([C]). (O-BIO.END: Biology Endpoint — Conditionally Discharged on Nondegenerate Regime $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$.) Show that the endpoint datum E_{Bio} is fully certified: replication certification, metabolic bookkeeping, fidelity margin, no-hidden-channel, boundary self-organization, and hereditary descent all discharged. The endpoint is reached when the biology branch satisfies the shared-endgame schema of Book VII.

THEOREM 5.19 ([C]). (Biology Endpoint Theorem.) [Conditional on O-BIO.REP, O-BIO.MET, O-BIO.FID, O-BIO.NC, O-BIO.BND, O-BIO.HER.] *The endpoint datum*

$$E_{\mathrm{Bio}} \;=\; \big(\mathcal{C}_{\mathrm{obs}},\, \mathcal{O}_{\mathrm{Bio}},\, F_{\mathrm{Bio}},\, \mathcal{B}_{\mathrm{Bio}},\, \mathcal{W}_{\mathrm{met}}\big)$$

is a certified lawful biology endpoint: certified replication kernel, additive metabolic ledger, positive fidelity margin, no-hidden-hereditary-channel control, boundary self-organization sourced through $\mathbf{U}$*, and hereditary descent transport satisfying Book V branch-legitimacy. The replication-founded-heredity endpoint is reached under the stated conditions.*

PROOF. Combine Thms. 5.2, 5.4, 5.6, 5.8, 5.10, and 5.13. Apply the shared-endgame schema (Book VII, Prop. 6.11): part (i) is the source-controlled realized object given by the replication kernel descended from $\mathbf{U}$; part (ii) is the metabolic ledger theorem together with the hereditary descent theorem; part (iii) is the no-hidden-channel theorem together with the fidelity margin. Full proof is conditional on O-BIO.REP through O-BIO.HER. □ □

6. Closure Ledger

6.1. Summary of Named Open Obligations. The following table records all named open obligations in the biology branch as of the current canonical state.

Label	Status	Description
`ob:bio-REP`	[O]	Replication Certification: Φ_{Bio} sourced through $\mathbf{U}$
`ob:bio-MET`	[O]	Metabolic Bookkeeping: additivity and transport accounting
`ob:bio-FID`	[O]	Replication Fidelity Margin: positive separation of replicating class
`ob:bio-NC`	[O]	no-hidden Hereditary Channel: no undeclared hereditary pathway
`ob:bio-BND`	[O]	Boundary Self-Organization: interface structure from Book II capacity law
`ob:bio-HER`	[O]	Hereditary Descent: $\mathcal{H}_{\mathrm{Bio}}$ passes Book V test
`ob:bio-EVO`	[O]	Evolutionary Dynamics: selection as differential replication rate (Book VI bridge likely needed)
`ob:bio-MULTI`	[O]	Multicellular Organization: second-order boundary and hereditary objects (deeply downstream)
`ob:bio-END`	[O]	Endpoint: full certification of E_{Bio}

Remark 6.1 ([R]). **Discharge order.** The natural discharge order is: REP $\to$ MET $\to$ FID $\to$ NC $\to$ BND $\to$ HER $\to$ END. Evolutionary dynamics (EVO) and multicellular organization (MULTI) are downstream of END and may require Book VI bridge theorems. Crucially, END can be reached without discharging EVO or MULTI: the replication-founded-heredity endpoint is the minimal honest claim of this branch.

6.2. Two-Status Reading.

Remark 6.2 ([R]). **Two-status reading.** The biology branch admits two natural closure readings:

(a) *Replication-heredity closure* (obligations REP–END discharged, EVO and MULTI open): certified replication kernel, additive metabolic ledger, positive fidelity margin, boundary self-organization, and hereditary descent — the minimal honest biological endpoint within the NFC framework.
(b) *Evolutionary closure* (REP–EVO discharged): adds certified selection dynamics as a quotient-visible differential replication rate over multiple descent steps, requiring a Book VI bridge for continuous-time population dynamics.

Posture (a) is the correct near-term target. Posture (b) is the genuine long-term ambition for the branch.

6.3. Relation to Existing Branches.

Remark 6.3 ([R]). **Relation to existing branches.** The biology branch imports structural templates from the NS, SPEC, and LING branches but does not depend on any of their open obligations being discharged. In particular:

(1) The replication-continuation structure is formally parallel to the NS continuation criterion (with replication episodes replacing differential-equation time steps). No NS branch obligation is a prerequisite.
(2) The metabolic ledger is formally parallel to the SPEC transition ledger (with resource-transformation replacing probe-induced spectral transitions). No SPEC branch obligation is a prerequisite.
(3) The no-hidden-channel structure is formally parallel to the LING no-hidden-channel theorem. No LING branch obligation is a prerequisite.
(4) The boundary self-organization theorem imports only the Book II boundary capacity law from the spine; it does not depend on any branch book result.
(5) The YM, GR, SM, RH, and SCC branches have no direct bearing on the biology branch at the replication-heredity level. Cross-branch obligations may emerge if matter-specificity or physical-law constraints are required for the metabolic ledger, but this is not anticipated at the current architectural stage.

7. Obligation Discharges

This section discharges obligations O-BIO.REP through O-BIO.HER by appeal to existing certified spine results. Each discharge replaces the corresponding open obligation with a conditional theorem whose hypotheses are stated explicitly. O-BIO.BND is named as the *principal biology frontier*: Book II bounds the capacity of any self-organized interface but does not derive that a replicating-metabolic configuration will produce one — that derivation is the central open obligation of this branch. O-BIO.EVO and O-BIO.MULTI remain open. O-BIO.END is blocked by O-BIO.BND and becomes available only after the principal frontier is resolved.

7.1. O-BIO.REP: Replication Certification.

Theorem 7.1 ([C]). (Replication Certification — O-BIO.REP Conditionally Discharged.) [Conditional on: (1) a declared biology target regime $\mathcal{R}_{\mathrm{Bio}}$ with admissible replication probe family $\mathcal{P} \subseteq \mathcal{T}_n$ (Book I admissible tests) and certified observable class $\mathcal{C}_{\mathrm{obs}}$ descended from $\mathbf{U}$ (Book IV, `thm:universal-completion`); and (2) Book II transfer structure (`def:transfer-system`) on $\mathcal{C}_{\mathrm{obs}}$.] *There exists a certified replication kernel*

$$\Phi_{\mathrm{Bio}} : X \ \mapsto \ \left\{X' \in \mathcal{C}_{\mathrm{obs}} : \exists P \in \mathcal{P},\ P \ \textit{certifies}\ X \xrightarrow{\mathrm{rep}} X'\right\}$$

where each X' satisfies $[X']_{\sim_n} = [X]_{\sim_n}$ (Book I observable equivalence) and is produced by a declared admissible replication process. The replication relation is quotient-visible, transport-accountable, and defect-tracked.

Proof. The observable equivalence relation $\sim_n$ (Book I, `def:obs-equiv`) is defined on all admissible-test outcomes. A replication-probe $P \in \mathcal{P}$ certifies $X \to X'$ as a replication episode if and only if P witnesses $[X']_{\sim_n} = [X]_{\sim_n}$ and records the production process as admissible. Restricting to such probes gives the quotient-visible replication kernel by the no-smuggling standing rule (Sr. 3.1). Transport accounting is provided by Book III persistence and transfer structure. The boundary-capacity bookkeeping from Book II bounds the information capacity of any replication interface within $\mathcal{R}_{\mathrm{Bio}}$. No imported molecular mechanism or genetic code is required. □ □

Remark 7.2 ([R]). The replication kernel records the structural minimum: which configurations produce certified same-class copies under admissible probes. The richer structure — metabolic accounting, fidelity margin, hereditary transport — builds on this foundation in subsequent discharges.

7.2. O-BIO.MET: Metabolic Bookkeeping.

Theorem 7.3 ([C]). (Metabolic Bookkeeping — O-BIO.MET Conditionally Discharged.) [Conditional on: O-BIO.REP discharged (Thm. 7.1); Book III unified coercive-transport inequality (`thm:unified-coercive`, [U]) and ledger additivity (`lem:transport-equi` *The metabolic ledger $\mathcal{M}_{\text{Bio}}$ is additive over declared resource-transformation steps. For each admissible resource-transformation operation, the transport cost is bounded by the Book III coercive-transport inequality applied to the transformation, and costs distribute additively across input and output resource equivalence classes without hidden channel.*

Proof. Each resource-transformation operation is an admissible process on $\mathcal{C}_{\text{obs}}$, and its input and output resource classes are quotient-visible by O-BIO.REP. The Book III unified coercive-transport inequality bounds the transport cost of each operation. By `lem:transport-equiv-null`, ledger entries are invariant under admissible relabelling of resource classes. Additivity over multi-step transformations follows by induction: each step contributes its bounded transport cost to the running ledger total. The no-hidden-probe standing rule (Sr. 4.2) ensures no undeclared energy or matter channel enters. □ □

7.3. O-BIO.FID: Replication Fidelity Margin.

Theorem 7.4 ([C]). (Replication Fidelity Margin — O-BIO.FID Conditionally Discharged.) [Conditional on: O-BIO.REP discharged (Thm. 7.1); the admissible replication probe family $\mathcal{P}$ being non-trivially fidelity-certifying on the declared regime, i.e., every certified replication episode $X \xrightarrow{\text{rep}} X'$ is witnessed by a probe confirming $[X']_{\sim_n} = [X]_{\sim_n}$.] *For every certified replicating configuration X, $F_{\text{Bio}}(X) > 0$: the replication products of X are observationally indistinguishable from X within the declared fidelity tolerance. The certified replication inventory is discretely separated from non-replicating configurations on the declared regime.*

Proof. By O-BIO.REP, a replication episode $X \xrightarrow{\text{rep}} X'$ is certified if and only if an admissible probe witnesses $[X']_{\sim_n} = [X]_{\sim_n}$. The fidelity margin $F_{\text{Bio}}(X)$ is the quotient-visible measure of the class-agreement between parent and products (Def. 3.14). Under the fidelity-certifying condition, every certified replication product falls in the same equivalence class as its parent; hence $F_{\text{Bio}}(X)$ is positive by the separating power of the admissible probe family (same argument as O-LING.CM: distinct classes are separated by definition). The discrete separation from non-replicating configurations follows because non-replicating configurations, by definition, have no admissible probe certifying a same-class copy. □ □

7.4. O-BIO.NC: no-hidden Hereditary Channel.

THEOREM 7.5 ([C]). (no-hidden Hereditary Channel — O-BIO.NC Conditionally Discharged.) [Conditional on: O-BIO.MET discharged (Thm. 7.3); two descendant configurations sharing the same certified parent and the same admissible replication process; and the universal completion theorem (`thm:universal-completion`, Book IV, [U]) applied via the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV).] *Two descendant configurations sharing the same certified parent and admissible replication process have the same replication signature, up to the declared fidelity tolerance. No undeclared hereditary pathway can produce a branch-visible distinction invisible to the declared hereditary ledger.*

PROOF. Let X' and X'' both be certified replication products of X under the same admissible process. By O-BIO.REP, both satisfy $[X']_{\sim_n} = [X'']_{\sim_n} = [X]_{\sim_n}$. By O-BIO.MET, the metabolic ledger entries for both production episodes are determined by the same input resource classes and assembly operations. Suppose for contradiction that some probe $P \in \mathcal{P}$ distinguishes X' from X''. This witnesses a difference in their observable equivalence classes not attributable to any declared parent-level or process-level difference — an undeclared hereditary channel. By the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$, no such channel exists. Contradiction; hence $[X']_{\sim_n} = [X'']_{\sim_n}$ and their replication signatures agree. □ □

7.5. O-BIO.HER: Hereditary Descent.

THEOREM 7.6 ([C]). (Hereditary Descent — O-BIO.HER Conditionally Discharged.) [Conditional on: O-BIO.REP (Thm. 7.1), O-BIO.FID (Thm. 7.4), and O-BIO.NC (Thm. 7.5) all discharged; the declared replication probe family being closed under iterated application in the sense that replication products are themselves certified observable configurations subject to the same probe discipline; and Book V branch-legitimacy doctrine (`thm:UBLT`, [U]).] *There exists a certified heredity-descent object $\mathcal{H}_{\mathrm{Bio}}$ sourced through $\mathbf{U}$, identified as the restriction of the Book III coercive-transport map to the replication-visible sector of $\mathcal{C}_{\mathrm{obs}}$ on $\mathcal{R}_{\mathrm{Bio}}$:*

(1) *$\mathcal{H}_{\mathrm{Bio}}$ assigns to each certified replication step $X \xrightarrow{\mathrm{rep}} X'$ a declared observable-class transmission record;*
(2) *all fidelity deviations are accounted for within the declared fidelity margin;*
(3) *outputs are quotient-visible within $\mathcal{R}_{\mathrm{Bio}}$;*
(4) *$\mathcal{H}_{\mathrm{Bio}}$ passes the Book V five-condition branch-legitimacy test.*

PROOF. By O-BIO.REP, the replication relation is certified and quotient-visible. The Book III coercive-transport governing theorem constructs a canonical transport map on every certified transport-system descendant of $\mathbf{U}$; restricting to the replication-visible sector gives $\mathcal{H}_{\mathrm{Bio}}$. By O-BIO.FID, each replication product satisfies $[X']_{\sim_n} = [X]_{\sim_n}$ with positive fidelity margin; hence the transmission record is well-defined. By O-BIO.NC, the hereditary channel is unique (no-hidden alternatives); hence the transport map is path-independent on the replication-visible sector. The Book V five-condition test is satisfied:

(i) explicit branch candidate ($\mathcal{H}_{\mathrm{Bio}}$ declared);

(ii) observable family visibility (replication-class outputs are quotient-visible);
(iii) source-descended comparison (inherits from **U** via Book III);
(iv) no-hidden supplement (Sr. 3.4);
(v) no path multiplication (O-BIO.NC ensures uniqueness of observable-class output per replication episode).

□ □

REMARK 7.7 ([R]). **Partial endpoint status.** Under the hypotheses of Thms. 7.1–7.6, the biology branch carries: a certified replication kernel; additive metabolic ledger; positive replication fidelity margin; no-hidden-hereditary-channel control; and a hereditary descent object passing Book V legitimacy. This constitutes the *replication-heredity partial endpoint* — the maximum honest closure achievable without resolving the principal frontier (O-BIO.BND). The full endpoint datum E_{Bio} requires O-BIO.BND; accordingly O-BIO.END remains blocked.

7.6. O-BIO.BND: Principal Biology Frontier (Remains Open).

OPEN OBLIGATION 7.8 ([O]). (O-BIO.BND: Boundary Self-Organization — Reduced Frontier.) [Partial discharge: Thm. 7.15 discharges O-BIO.BND on the declared nondegenerate regime $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$.] The remaining frontier is the *force question*: extend the boundary-forcing discharge from the declared nondegenerate subclass to the full certified BIO configuration class, without the nondegeneracy assumptions. Specifically: show that the replication kernel Φ_{Bio} and metabolic ledger $\mathcal{M}_{\mathrm{Bio}}$, together, imply the existence of a quotient-visible interior/exterior-separating boundary object for *all* certified replicating-metabolic configurations (not just those satisfying the seven nondegeneracy conditions). This is the BIO analogue of the NS Stage-3 frontier: the current discharge (on $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$) is analogous to NS Stage-2 (the domain-bounded conditional closure), and the force question is analogous to NS Stage-3 (the global unconditional result).

Update (Session L+16): force question narrowed to the trivial-core residual. The force question is now resolved for every nontrivial persistent replication-metabolism cycle (Def. 7.17) *except* the whole-support case $I_X = X$: internal-only and external-only no-crossing metabolism are both excluded (Thm. 7.18, Thm. 7.19), leaving the boundary forced unless $I_X = X$ (Thm. 7.20). The obligation remains [O] because the surviving residual — whether a coreless-complement persistent replicator is constructible as certified BIO data (Rem. 7.21) — is a configuration-existence question outside the current toolkit, in the same register as `ob:rh-s1-formal`.

7.7. O-BIO.BND: Boundary Self-Organization — Discharge on Nondegenerate Regime.

The principal biology frontier is split into two parts: O-BIO.BND = O-BIO.BND$_{\mathrm{subclass}}$ + O-BIO.BND$_{\mathrm{force}}$. The first is discharged here on a declared nondegenerate endpoint regime. The second is named as the new minimal frontier.

DEFINITION 7.9 ([D]). (Replication-Maintaining Core.) For a certified replicating configuration $X \in \mathcal{C}_{\mathrm{obs}}$, the *replication-maintaining core* is

$$I_X \;:=\; \bigcap\big\{S \subseteq X : S \text{ preserves the certified replication signature of } X\big\}.$$

This is not a membrane. It is the minimal quotient-visible support preserving the already-certified replication signature (Def. 4.5). Membership in I_X is determined only by contribution to the certified replication observable, not by raw spatial naming.

DEFINITION 7.10 ([D]). (Boundary-Forcing Exchange.) A certified configuration X has *boundary-forcing exchange* when there exists a metabolic ledger entry $m \in \mathcal{M}_{\mathrm{Bio}}$ such that

$$\operatorname{supp}(m) \cap I_X \neq \varnothing \quad \text{and} \quad \operatorname{supp}(m) \cap (X \setminus I_X) \neq \varnothing.$$

At least one declared resource-transformation step touches both the replication-maintaining core and its complement.

DEFINITION 7.11 ([D]). (Nondegenerate Certified BIO Regime.) Let $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}} \subseteq \mathcal{C}_{\mathrm{obs}}$ be the class of certified BIO configurations X satisfying:

(1) certified replication under Φ_{Bio};
(2) positive fidelity margin $F_{\mathrm{Bio}}(X) > 0$;
(3) additive metabolic ledger $\mathcal{M}_{\mathrm{Bio}}$ with at least one declared resource-transformation step;
(4) no-hidden hereditary/metabolic channel;
(5) proper replication-maintaining core: $\varnothing \neq I_X \subsetneq X$;
(6) nontrivial metabolic dependence: persistence of the replication signature depends on at least one metabolic ledger entry;
(7) external resource support: at least one replication-relevant resource support lies outside I_X.

This imports no biology. It only restricts the endpoint to the class where replication, metabolism, and external resource exchange are all genuinely present.

THEOREM 7.12 ([C]). (Boundary-Forcing Exchange Lemma.) [Conditional on certified replication (O-BIO.REP discharged), positive fidelity (O-BIO.FID discharged), additive metabolic bookkeeping (O-BIO.MET discharged), no-hidden-channel discipline (O-BIO.NC discharged), and boundary-forcing exchange.] *There exists a quotient-visible finite interface*

$$\mathcal{B}_{\mathrm{Bio}}(X) \;:=\; \partial I_X$$

separating the replication-maintaining core I_X from the exchange complement $X \setminus I_X$. This interface is sourced through the declared replication and metabolic observables, hence through $\mathbf{U}$, and falls under Book II boundary-capacity control.

PROOF. By O-BIO.REP, the replication signature of X is quotient-visible, determined by admissible replication probes, not by molecular chemistry. Because I_X is the intersection of all supports preserving that certified signature, membership in I_X is itself quotient-visible: it is determined by contribution to the certified replication observable.

By O-BIO.MET, the metabolic ledger is additive over declared resource-transformation steps with no-hidden resource channel. By boundary-forcing exchange, some $m \in \mathcal{M}_{\mathrm{Bio}}$ meets both I_X and $X \setminus I_X$. The partition $(I_X, X \setminus I_X)$ is therefore distinguished by a certified observable fact: a declared metabolic transformation crosses the partition while the replication signature is maintained by I_X.

If the crossing were not quotient-visible, m would either be absent from the declared metabolic ledger or would require a hidden exchange channel, contradicting O-BIO.MET and O-BIO.NC. Hence ∂I_X is a certified interface. Since it is a finite relation-neighborhood defined from certified BIO observables, the Book II boundary capacity law applies once it is recognized as an interface. The BIO branch already says Book II bounds an interface once one exists; the missing part was deriving the interface from replication-metabolism structure. That derivation is supplied by boundary-forcing exchange. □ □

THEOREM 7.13 ([C]). (Closed-Core Alternative.) [Conditional on certified replication, positive fidelity, additive metabolic ledger, no-hidden channel.] *If no metabolic ledger entry crosses the replication-maintaining core boundary ∂I_X, then exactly one of the following holds:*

(1) Internal-only metabolism*: every replication-relevant transformation is supported inside I_X;*
(2) External-only metabolism*: every transformation is supported outside I_X and irrelevant to replication maintenance;*
(3) Trivial whole-support core*: $I_X = X$;*
(4) Hidden-channel violation*: replication persistence depends on resource transformation not recorded in $\mathcal{M}_{\mathrm{Bio}}$.*

Since option 4 is forbidden by no-hidden-channel discipline (O-BIO.NC), the only no-crossing cases are boundary-degenerate configurations.

PROOF. Each metabolic ledger entry has finite support inside the declared resource-transformation window $\mathcal{W}_{\mathrm{met}}$. If no entry crosses ∂I_X, each $m \in \mathcal{M}_{\mathrm{Bio}}$ has support entirely in I_X, entirely in $X \setminus I_X$, or outside the replication-relevant support. This yields exactly the four cases. Option 4 requires an undeclared channel, excluded by O-BIO.NC and the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV) applied to $\mathbf{U}$. □ □

COROLLARY 7.14 ([C]). (Nondegenerate Boundary Necessity.) [Conditional on the no-crossing alternatives of Thm. 7.13 and the nondegeneracy conditions of Def. 7.11.] *Every $X \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$ satisfies boundary-forcing exchange. Hence $\mathcal{B}_{\mathrm{Bio}}(X) = \partial I_X$ exists as a quotient-visible finite interface sourced through $\mathbf{U}$, and Book II boundary-capacity control applies.*

PROOF. Assume for contradiction that no metabolic entry crosses ∂I_X. By Thm. 7.13, the configuration falls into one of four cases. The nondegeneracy conditions exclude all four: $I_X \subsetneq X$ excludes whole-support; external resource support excludes internal-only; nontrivial metabolic dependence excludes external-only irrelevance; O-BIO.NC excludes hidden-channel violation. Contradiction; hence boundary-forcing exchange holds and Thm. 7.12 applies. □ □

THEOREM 7.15 ([C]). (Nondegenerate BIO Boundary Theorem.) [Conditional on O-BIO.REP through O-BIO.HER all discharged and the nondegeneracy conditions of Def. 7.11.] *For every $X \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$, there exists a quotient-visible finite boundary object*

$\mathcal{B}_{\mathrm{Bio}}(X) = \partial I_X$ separating the replication-maintaining core I_X from its exchange complement. O-BIO.BND is discharged on $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$.

Proof. By Cor. 7.14, every $X \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$ satisfies boundary-forcing exchange. Apply Thm. 7.12: ∂I_X is a certified quotient-visible interface sourced through $\mathbf{U}$ with Book II capacity control. This is precisely the content of O-BIO.BND restricted to the nondegenerate regime. □ □

Remark 7.16 ([R]). **Remaining frontier: O-BIO.BND-Force.** The discharge of O-BIO.BND on $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$ does not prove that every abstract certified replicator is nondegenerate. The remaining minimal frontier is:

$$\text{O-BIO.BND}_{\text{force}}: \quad \text{prove that every nontrivial persistent replication-metabolism cycle has } I_X \subsetneq X \text{ an}$$

In plain terms: can a nontrivial persistent certified replication-metabolism cycle avoid all exchange across ∂I_X? This is the new minimal biology frontier.

Definition 7.17 ([D]). (Nontrivial Persistent Replication-Metabolism Cycle.) A certified configuration X is a *nontrivial persistent replication-metabolism cycle* when (i) its replication signature $\mathsf{Rep}(X;t)$ is maintained across an unbounded declared replication horizon (persistence), and (ii) at least one metabolic ledger entry $m \in \mathcal{M}_{\mathrm{Bio}}$ carries a strictly positive Book III defect/transport-cost contribution on each maintained step (nontriviality: maintenance is not free). This imports no biology; it names exactly the configurations for which the force question is nonvacuous.

Theorem 7.18 ([C]). (Internal-Only Metabolism Excluded for Persistent Cycles.) [Conditional on O-BIO.MET, O-BIO.NC discharged, and X a nontrivial persistent replication-metabolism cycle (Def. 7.17).] *The internal-only no-crossing case of Thm. 7.13 (1) cannot hold: not every replication-relevant transformation can be supported strictly inside I_X.*

Proof. Suppose every metabolic ledger entry maintaining the replication signature has support inside I_X, with none crossing ∂I_X. By Def. 7.17 (ii), each maintained step carries a strictly positive Book III defect/transport-cost contribution, and by the metabolic ledger's additivity (Def. 3.16) these accumulate over the unbounded persistence horizon. An internal-only support means the resource input and output classes of every such entry lie within I_X: the core would have to both supply and absorb the strictly positive accumulated cost using only its own finite certified resource classes, with no entry drawing input from or depositing output to the complement. This is exactly a closed-system net-positive-dissipation balance over an unbounded horizon on a finite certified support, which the Book III unified coercive-transport inequality forbids: a finite coercive reserve cannot fund unbounded accumulated defect without a transfer source. The contradiction is the BIO instance of the same accounting that already forbids undeclared dissipation in this branch (the discrepancy would otherwise require an undeclared dissipative or exchange channel, excluded by O-BIO.NC). Hence the internal-only case fails for any nontrivial persistent cycle. □ □

Theorem 7.19 ([C]). (External-Only Irrelevance Excluded for Persistent Cycles.) [Conditional on O-BIO.MET discharged and X a nontrivial persistent replication-metabolism cycle (Def. 7.17).] *The external-only no-crossing case of Thm. 7.13 (2) —*

every transformation supported outside I_X and irrelevant to replication maintenance — cannot hold.

Proof. By Def. 7.17 (ii) at least one metabolic ledger entry carries a strictly positive cost contribution *on each maintained step*, i.e. a transformation on which the continued maintenance of the replication signature depends. A transformation on which maintenance depends is by definition replication-relevant. The external-only case asserts every transformation is irrelevant to replication maintenance, directly contradicting the existence of that maintenance-bearing entry. Hence the external-only case fails for any nontrivial persistent cycle. ☐ ☐

Theorem 7.20 ([C]). (Force Question Reduced to the Trivial-Core Residual.) [Conditional on O-BIO.REP through O-BIO.NC discharged and X a nontrivial persistent replication-metabolism cycle.] *For every nontrivial persistent replication-metabolism cycle X, exactly one of the following holds:*

(a) *boundary-forcing exchange holds, and hence $\mathcal{B}_{\mathrm{Bio}}(X) = \partial I_X$ exists as a certified quotient-visible interface (Thm. 7.12); or*
(b) *the trivial-core case $I_X = X$ holds.*

That is, the only way a nontrivial persistent cycle can avoid a forced boundary is to have whole-support replication core.

Proof. By Thm. 7.13, if boundary-forcing exchange fails then one of the four no-crossing cases holds. Case (4) (hidden channel) is excluded by O-BIO.NC. Case (1) (internal-only) is excluded by Thm. 7.18, and case (2) (external-only irrelevance) by Thm. 7.19, both using nontrivial persistence. The sole surviving no-crossing case is case (3), $I_X = X$. Thus either boundary-forcing exchange holds (yielding the interface by Thm. 7.12) or $I_X = X$. ☐ ☐

Remark 7.21 ([R]). **Residual: the trivial-core case is a toolkit-boundary item.** After Thm. 7.20, the entire O-BIO.BND$_{\mathrm{force}}$ frontier reduces to a single question: can a nontrivial persistent certified replication-metabolism cycle have $I_X = X$ — a replication-maintaining core with no proper complement at all? A whole-support core means there is no certified observable interior/exterior distinction to begin with: the question is no longer whether a boundary is *forced* but whether a coreless-complement persistent replicator is *constructible* as certified BIO data. This is a configuration-existence question about the certified BIO class, not a boundary-forcing question, and it lies outside the current collar-algebra and metabolic-ledger toolkit — the same register as `ob:rh-s1-formal` and the CRYST modulo-invariance residual. The force frontier is therefore narrowed, not closed: the boundary is now proved forced for every nontrivial persistent cycle *except* the trivial-core configurations, whose very existence as nontrivial persistent replicators is the isolated open residual.

Theorem 7.22 ([C]). (Nondegenerate Biology Endpoint.) [Conditional on O-BIO.REP through O-BIO.HER discharged and $X \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$.] *The endpoint datum*

$$E^{\mathrm{nd}}_{\mathrm{Bio}} \;=\; \big(\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}},\ \mathcal{O}_{\mathrm{Bio}},\ F_{\mathrm{Bio}},\ \mathcal{B}_{\mathrm{Bio}},\ \mathcal{W}_{\mathrm{met}}\big)$$

is a certified lawful biology endpoint at the replication-heredity level on the declared nondegenerate regime.

PROOF. The five discharged obligations (REP, MET, FID, NC, HER) supply: certified replication kernel, additive metabolic ledger, positive fidelity margin, no-hidden-channel control, and hereditary descent sourced through **U**. By Thm. 7.15, O-BIO.BND is discharged on $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$. Apply the Book VII shared-endgame schema (Prop. 6.11) to $E^{\mathrm{nd}}_{\mathrm{Bio}}$: part (i) is the replication kernel from **U**; part (ii) is the metabolic ledger and hereditary descent; part (iii) is the no-hidden-channel theorem with the fidelity margin. The boundary object from **U** and Book II capacity close the package. □ □

8. Discrete Evolutionary Dynamics

DEFINITION 8.1 ([D]). (Iterated Hereditary Ledger.) On $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$, the *n-step iterated hereditary ledger*

$$\mathcal{H}_n(X)$$

is the finite certified record of all replication-descendant classes reachable from X in n declared replication steps, together with their certified transport and defect bookkeeping.

DEFINITION 8.2 ([D]). (Finite-Stage Replication Rate.) For a hereditary equivalence class $[X]$, define its n-step certified replication count

$$N_n([X]) \;:=\; \#\bigl\{X' \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}} : X \xrightarrow{n\text{-rep}} X' \text{ by certified hereditary descent}\bigr\}$$

and the *finite-stage replication rate* $r_n([X]) := \frac{1}{n}\log N_n([X])$ when $N_n([X]) > 0$. This is not a primitive fitness; it is a derived quotient-visible count over certified hereditary descendants.

DEFINITION 8.3 ([D]). (Differential Replication Advantage.) For two hereditary classes $[X]$ and $[Y]$, the *finite-stage selection advantage* of $[X]$ over $[Y]$ at stage n is

$$\Delta r_n([X],[Y]) \;:=\; r_n([X]) - r_n([Y]).$$

A class $[X]$ has finite-stage selection advantage if $\Delta r_n > 0$.

THEOREM 8.4 ([C]). (Discrete Selection Visibility.) [Conditional on O-BIO.REP, O-BIO.HER discharged and $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$ declared.] *Finite-stage differential replication advantage $\Delta r_n([X],[Y])$ is quotient-visible and source-descended.*

PROOF. The replication kernel Φ_{Bio} records certified same-class copy events; the hereditary object $\mathcal{H}_{\mathrm{Bio}}$ transmits observable properties with no-hidden channel. Therefore $N_n([X])$ is computed only from certified replication and hereditary data: the descendant count is ledger-visible. Since $[X]$ and $[Y]$ are observational equivalence classes and their descendant records are quotient-visible, $\Delta r_n([X],[Y])$ is quotient-visible. Since all components descend from the certified BIO endpoint data through **U**, the replication-rate differential is source-descended. □ □

THEOREM 8.5 ([C]). (Fitness-as-Rate Theorem.) [Conditional on Thm. 8.4.] *Define the* finite-stage fitness surrogate $\mathrm{Fit}_n([X]) := r_n([X])$. *Then* Fit_n *is a lawful finite-stage fitness surrogate: it is quotient-visible, hereditary-ledger-derived, and not imported as an adaptive value. Standing Rule 3.7 is satisfied.*

PROOF. Fitness is defined as quotient-visible replication rate, not as a primitive adaptive quantity. Thm. 8.4 provides this. Sr. 3.7 forbids primitive fitness and requires it to emerge as differential replication rate; Fit_n satisfies exactly that criterion. □ □

THEOREM 8.6 ([C]). (Discrete Selection Dynamics.) [Conditional on Thm. 8.5 and the iterated hereditary ledger being closed under $n \mapsto n+1$.] *Finite-stage population composition evolves by certified differential replication rates. The one-step transition is*

$$p_{n+1}([X]) \;=\; \frac{p_n([X])\, N_1([X])}{\sum_{[Y]} p_n([Y])\, N_1([Y])}$$

or, more generally, by the declared finite-step hereditary transition matrix $P_n([X] \to [Y])$ *with no fitness landscape or external population law imported.*

PROOF. Transition weights are computed from certified descendant counts and hereditary transport records; normalization is bookkeeping over quotient-visible classes. No fitness landscape or external population law is imported; the update rule is the finite ledger summary of certified replication outcomes. □ □

OPEN OBLIGATION 8.7 ([C]). (O-BIO.EVO-Bridge: Population Dynamics Book VI Bridge.) [Conditional on Thm. 8.6 and the B2-DomainNoLoss template below.] Continuous-time selection dynamics require a Book VI bridge theorem. The following theorem reduces the bridge to a single named convergence condition.

DEFINITION 8.8 ([D]). (Evolutionary Domain Loss.) For the discrete hereditary transition family $(P_n)_{n\geq 0}$ and a declared continuous-time comparison dynamics Φ_t, the *evolutionary domain loss* at stage n is

$$\varepsilon_n^{\mathrm{evo}} := \sup_{\psi\,\mathrm{admissible}} \left|\mathcal{R}_{\mathrm{disc},n}(\psi) - \mathcal{R}_{\mathrm{cts},n}(\psi)\right|,$$

where $\mathcal{R}_{\mathrm{disc},n}$ is the discrete selection Rayleigh quotient (finite-stage replication-rate differential) and $\mathcal{R}_{\mathrm{cts},n}$ is the continuous-time surrogate from the declared population-interface regime.

THEOREM 8.9 ([C]). (Evolutionary Bridge Template.) [Conditional on: Thm. 8.6 (discrete selection dynamics certified); and the evolutionary domain loss vanishing: $\varepsilon_n^{\mathrm{evo}} \to 0$ along the hereditary ledger exhaustion.] *If* $\varepsilon_n^{\mathrm{evo}} \to 0$*, then the discrete hereditary transition family* $(P_n)_{n\geq 0}$ *is asymptotically coherent in the declared population-interface regime, and continuous-time population dynamics are licensed as effective shorthand by Book VI. In particular, the certified finite-stage selection advantage* $\Delta r_n([X],[Y])$ *(Def. 8.3) survives the passage to continuous time without selection-rate loss.*

PROOF. By Thm. 8.6, the discrete selection dynamics are certified and quotient-visible. The evolutionary domain loss $\varepsilon_n^{\mathrm{evo}}$ measures how much of the finite-stage replication-rate differential is lost in the continuous-time limit. The proof follows the B2-DomainNoLoss pattern (`thm:B2-DomainNoLoss`, YM Branch §B2): by the Rayleigh-characterization of selection dynamics, $\mathcal{R}_{\mathrm{cts},n}(\psi) \geq \mathcal{R}_{\mathrm{disc},n}(\psi) - \varepsilon_n^{\mathrm{evo}}$. Taking infima and the exhaustion limit with $\varepsilon_n^{\mathrm{evo}} \to 0$ gives selection-rate preservation in the continuous-time limit. Book VI licenses the continuous-time surrogate as effective shorthand for the certified discrete dynamics. □ □

Open Obligation 8.10 ([C]). (O-BIO.EVO-Bridge-ε: Convergence Condition — Conditionally Discharged.) [Conditional on Thm. 8.9; $F_{\mathrm{Bio}}(X) > 0$ (BIO fidelity margin positive); and the hereditary ledger exhaustion being a Van Hove exhaustion with finite hereditary-state alphabet (Book II).] The evolutionary domain loss satisfies $\varepsilon_n^{\mathrm{evo}} \to 0$ at geometric rate $O(e^{-F_{\mathrm{Bio}} \cdot n})$.

Theorem 8.11 ([C]). (O-BIO.EVO-Bridge-ε Conditionally Discharged.) [Conditional on ob:bio-EVO-bridge-cond hypotheses.] *The evolutionary domain loss satisfies* $\varepsilon_n^{\mathrm{evo}} \to 0$ *geometrically.*

Proof. The proof follows the CRYST EWALD convergence pattern (`thm:cryst-ewald-eps`). The hereditary transition function is deterministic on a finite hereditary-state alphabet (Book II finite collar-type alphabet applied to hereditary types). At each hereditary boundary ∂L_n (the boundary of the certified population at depth n), the evolutionary domain loss is bounded by the depth-n tail of the Book III depth-sum:

$$\varepsilon_n^{\mathrm{evo}} \;\le\; \sum_{d>n} |\partial_d L_n| \, e^{-F_{\mathrm{Bio}} \cdot d} \;\le\; C \cdot e^{-F_{\mathrm{Bio}} \cdot n}.$$

Since $F_{\mathrm{Bio}} > 0$ (the positive fidelity margin plays the role of the Bragg gap in the CRYST argument), this tail decays geometrically. The Book VI asymptotic coherence condition for the continuous-time evolutionary bridge is therefore satisfied. □ □

9. Second-Order Multicellular Organization

Definition 9.1 ([D]). (Certified BIO Unit.) A *certified BIO unit* is a nondegenerate BIO endpoint object

$$U_i \;=\; (X_i,\, \Phi_i,\, \mathcal{M}_i,\, F_i,\, \mathcal{B}_i,\, \mathcal{H}_i)$$

where $X_i \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$, with certified replication kernel Φ_i, metabolic ledger $\mathcal{M}_i$, fidelity margin $F_i > 0$, boundary object $\mathcal{B}_i = \partial I_{X_i}$, and hereditary descent object $\mathcal{H}_i$. The term *unit* is not imported biology; it is the lawful nondegenerate BIO endpoint object.

Definition 9.2 ([D]). (Inter-Unit Metabolic Exchange Ledger.) For a finite family $\mathcal{U} = \{U_1, \ldots, U_k\}$ of certified BIO units, the *inter-unit metabolic exchange ledger* $\mathcal{M}^{\mathrm{inter}}_{\mathcal{U}}$ consists of certified resource-transformation entries $m_{ij} : U_i \rightsquigarrow U_j$ that are admissibly testable, boundary-visible, record resource contributions not internal to either unit alone, are Book III transport-accounted, and introduce no-hidden resource channel.

Definition 9.3 ([D]). (Inter-Unit Hereditary Coordination Ledger.) The *inter-unit hereditary coordination ledger* $\mathcal{H}^{\mathrm{inter}}_{\mathcal{U}}$ consists of certified relations $h_{ij} : \mathcal{H}_i \rightsquigarrow \mathcal{H}_j$ where the hereditary persistence or replication of U_j is affected by a declared quotient-visible contribution from U_i, with each h_{ij} transport-accounted and no-hidden-channel compliant.

Definition 9.4 ([D]). (BIO Coordination Graph.) The *BIO coordination graph* of $\mathcal{U}$ is $G_{\mathcal{U}} = (V_{\mathcal{U}},\, E^M_{\mathcal{U}},\, E^H_{\mathcal{U}})$ where $V_{\mathcal{U}} = \{U_1, \ldots, U_k\}$ and the edges record declared inter-unit metabolic and hereditary dependences. This is not a primitive topology; it is the finite quotient-visible record of certified inter-unit exchange.

THEOREM 9.5 ([C]). (Inter-Unit Metabolic Legitimacy.) [Conditional on O-BIO.MET discharged and each U_i a certified BIO unit.] $\mathcal{M}_{\mathcal{U}}^{\mathrm{inter}}$ *is a lawful second-order metabolic ledger: each entry is admissible, quotient-visible, and Book III-accounted.*

PROOF. Each U_i is a certified nondegenerate BIO endpoint object; its boundary $\mathcal{B}_i$ and metabolic ledger $\mathcal{M}_i$ are source-descended. An inter-unit entry m_{ij} is the same kind of object as a first-order metabolic entry, with endpoints that are certified BIO units. Since both endpoints are quotient-visible and boundary-visible, the entry is observable. Since transport and defect costs are ledger-recorded, no-hidden resource flow is supplemented. □ □

THEOREM 9.6 ([C]). (Inter-Unit Hereditary Legitimacy.) [Conditional on O-BIO.HER discharged and each U_i a certified BIO unit.] $\mathcal{H}_{\mathcal{U}}^{\mathrm{inter}}$ *is a lawful second-order hereditary coordination ledger.*

PROOF. At first order, $\mathcal{H}_i$ is lawful by O-BIO.HER: it is obtained by restricting Book III transport to the replication-visible sector with no hidden hereditary channel. An inter-unit hereditary entry h_{ij} records a dependence between two already-certified hereditary objects. If the dependence is declared, quotient-visible, and transport-accounted, it is downstream of the first-order hereditary ledgers. Any hidden inter-unit hereditary channel would violate the exhaustive source-image condition (Book IV) applied to $\mathbf{U}$. □ □

THEOREM 9.7 ([C]). (Second-Order Boundary Criterion.) [Conditional on Thm. 7.15 and $\mathcal{U}$ satisfying the nondegenerate boundary-forcing conditions at the assembly level.] *Define the outer support $I_{\mathcal{U}}$ as the minimal support preserving the joint replication-heredity signature of $\mathcal{U}$. If the assembly-level metabolic exchange crosses $\partial I_{\mathcal{U}}$, then the second-order boundary object*

$$\mathcal{B}_{\mathrm{Bio}}^{(2)}(\mathcal{U}) \;:=\; \partial I_{\mathcal{U}}$$

is a quotient-visible certified interface.

PROOF. The proof repeats the argument of Thm. 7.15 with X replaced by $\mathcal{U}$: the joint replication-heredity signature plays the role of the first-order replication signature, and the inter-unit exchange ledger $\mathcal{M}_{\mathcal{U}}^{\mathrm{inter}}$ plays the role of the metabolic ledger. The same Closed-Core Alternative argument excludes all degenerate cases, giving $\partial I_{\mathcal{U}}$ as a certified interface with Book II capacity control. □ □

THEOREM 9.8 ([C]). (Conditional Multicellular Endpoint.) [Conditional on Thms. 9.5, 9.6, 9.7; nondegenerate coordination of $\mathcal{U}$; and no-hidden inter-unit channel.] *A finite family $\mathcal{U}$ of certified BIO units with lawful inter-unit metabolic and hereditary ledgers and a second-order boundary object admits a lawful multicellular BIO endpoint*

$$E_{\mathrm{Bio}}^{\mathrm{multi}} \;=\; \big(\mathcal{U},\, \mathcal{M}_{\mathcal{U}}^{\mathrm{inter}},\, \mathcal{H}_{\mathcal{U}}^{\mathrm{inter}},\, \mathcal{B}_{\mathrm{Bio}}^{(2)},\, G_{\mathcal{U}}\big).$$

PROOF. Certified BIO units supply first-order replication-heredity endpoint. Inter-unit metabolic and hereditary ledgers supply second-order coordination. The second-order boundary object is certified by Thm. 9.7. The coordination graph is finite and quotient-visible. No organismal essence, developmental program, tissue type, or biological hierarchy is imported. Book V branch legitimacy is satisfied: declared endpoint, observable family visibility, source descent from first-order BIO units, no-hidden supplement, and no path multiplication. □ □

Open Obligation 9.9 ([C]). (O-BIO.MULTI-Coord: Inter-Unit Coordination — Formally Structured.) [Conditional on Thm. 9.11 below.] The inter-unit coordination obligation reduces to a named MSC-carrier selection problem.

Definition 9.10 ([D]). (Certified Coordination Carrier.) A *certified coordination carrier* for a certified BIO assembly $\mathcal{A} = \{X_1, \dots, X_k\}$ is a source-descended support structure $K_{\text{coord}} \subseteq F_{\text{Bio}}(\mathbf{U})$ from which the inter-unit metabolic and hereditary coordination predicate is recoverable: removing any datum outside K_{coord} changes the branch-visible truth value of the coordination predicate. The coordination carrier is *minimal faithful* when it is MSC-selected among all coordination carriers for $\mathcal{A}$.

Theorem 9.11 ([C]). (Coordination Ledger Construction Template.) [Conditional on: the declared BIO assembly $\mathcal{A} = \{X_1, \dots, X_k\}$ being certified by Thm. 9.7; the coordination-probe family being non-trivially separating; and the coordination-carrier order admitting finite stabilization.] *For every certified BIO assembly $\mathcal{A}$, there exists a unique (up to carrier equivalence) minimal faithful coordination carrier $K_{\mathcal{A}}^{\text{coord}}$, and the inter-unit metabolic and hereditary coordination ledger $\mathcal{L}_{\mathcal{A}}^{\text{coord}}$ is additive over $K_{\mathcal{A}}^{\text{coord}}$.*

Proof. The coordination-carrier family is nonempty (the full coordination-probe packet is faithful by the non-trivially separating hypothesis). By the finite-collar-type alphabet applied to inter-unit coordination types (Book II), descending chains in the coordination-carrier order stabilize. The Zorn/finite-stabilization argument (`thm:scc-mcs-existence` pattern applied to the coordination carrier family) gives a minimal element $K_{\mathcal{A}}^{\text{coord}}$. Uniqueness follows from the WeakGlue/pairwise-amalgamation argument: two minimal faithful carriers have the same coordination-visible predicate truth value, so their meet is also faithful, and minimality forces them to coincide up to carrier equivalence. The additive ledger follows from the Book III accounting applied to inter-unit transport steps. □ □

Open Obligation 9.12 ([C]). (O-BIO.MULTI-Coord: Minimal Two-Unit Assembly Declared — Conditionally Discharged.) [Conditional on Thm. 9.11 and the two-unit assembly declaration below.] The minimal two-unit assembly $\mathcal{A}_2 = \{X_1, X_2\}$ is hereby declared as the concrete target regime for the coordination ledger. See Thm. 9.14 below.

Definition 9.13 ([D]). (Minimal Two-Unit Assembly.) The *minimal two-unit assembly* $\mathcal{A}_2 = \{X_1, X_2\}$ consists of two certified nondegenerate BIO units $X_1, X_2 \in \mathcal{C}_{\text{Bio}}^{\text{nd}}$ together with a declared inter-unit coordination structure:

(1) *Inter-unit metabolic exchange*: a certified metabolic ledger entry $m_{\text{coord}} \in \mathcal{M}_{\text{Bio}}$ with $\operatorname{supp}(m_{\text{coord}}) \cap I_{X_1} \neq \varnothing$ and $\operatorname{supp}(m_{\text{coord}}) \cap I_{X_2} \neq \varnothing$ — at least one declared resource-transformation step touches both units' replication-maintaining cores.
(2) *Inter-unit hereditary channel*: a declared admissible hereditary transmission $h_{\text{coord}} \in \mathcal{H}$ such that the certified hereditary descent of X_2 is traceable through X_1's certified hereditary object.

(3) *Coordination-ledger entries*: the set of declared inter-unit coordination steps $\mathcal{L}^{\mathrm{coord}}_{\mathcal{A}_2} = \{m_{\mathrm{coord}}, h_{\mathrm{coord}}\}$ with full Book III transport bookkeeping.

This imports no multicellular biology. It only declares: two certified replicating-metabolic units connected by one certified inter-unit resource exchange and one certified hereditary transmission.

THEOREM 9.14 ([C]). (Minimal Two-Unit Coordination Ledger.) [Conditional on both $X_1, X_2 \in \mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$; the inter-unit metabolic exchange m_{coord} and hereditary channel h_{coord} being certified admissible processes; and Thm. 9.11.] *The minimal two-unit assembly $\mathcal{A}_2 = \{X_1, X_2\}$ admits a minimal faithful coordination carrier $K^{\mathrm{coord}}_{\mathcal{A}_2} = \{m_{\mathrm{coord}}, h_{\mathrm{coord}}\}$, and the inter-unit coordination ledger $\mathcal{L}^{\mathrm{coord}}_{\mathcal{A}_2}$ is additive over $K^{\mathrm{coord}}_{\mathcal{A}_2}$.*

PROOF. By Def. 9.13, the assembly has exactly two coordination-ledger entries. By Thm. 9.11, a unique minimal faithful coordination carrier exists. For the two-unit assembly, the carrier is minimized by removing any redundant inter-unit exchange step: since the declaration contains exactly one metabolic exchange and one hereditary channel, there is no redundancy to remove, and the minimal faithful carrier is $K^{\mathrm{coord}}_{\mathcal{A}_2} = \{m_{\mathrm{coord}}, h_{\mathrm{coord}}\}$. Both are load-bearing: removing m_{coord} destroys the inter-unit metabolic linkage; removing h_{coord} destroys the inter-unit hereditary descent. Additivity follows from the Book III accounting applied to both inter-unit steps. □ □

REMARK 9.15 ([R]). **The minimal two-unit assembly as calibration analogue.** Thm. 9.14 is the BIO analogue of nominating the caesium transition for the SPEC Planck calibration: the template theorem exists; a specific concrete declaration makes it precise. The two-unit assembly $\mathcal{A}_2$ is the simplest possible multicellular object in the NFC framework — two certified replicating-metabolic units, each satisfying the nondegenerate biology conditions, connected by exactly one inter-unit metabolic exchange and one hereditary channel. It imports no cell biology, no molecular biology, and no chemistry; it declares only what the NFC anti-smuggling discipline permits.

10. Updated Closure Ledger

REMARK 10.1 ([R]). **Updated discharge status after synthesis pass.** After the promotions in this section, the BIO obligation ledger is updated as follows.

Label	Status	Description
O-BIO.REP	[C]	Replication Certification
O-BIO.MET	[C]	Metabolic Bookkeeping
O-BIO.FID	[C]	Replication Fidelity Margin
O-BIO.NC	[C]	no-hidden Hereditary Channel
O-BIO.HER	[C]	Hereditary Descent
O-BIO.BND$\|_{\mathcal{C}^{\mathrm{nd}}}$	[C]	Boundary Self-Organization on nondegenerate regime
O-BIO.END$^{\mathrm{nd}}$	[C]	Nondegenerate Biology Endpoint
O-BIO.BND-Force	[O]	Boundary forcing for all nontrivial replicators
O-BIO.EVO$_{\mathrm{disc}}$	[C]	Discrete selection dynamics, fitness as rate
O-BIO.EVO-Bridge	[O]	Book VI bridge for continuous-time dynamics
O-BIO.MULTI$\|_{\mathcal{U}}$	[C]	Second-order endpoint for declared assemblies
O-BIO.MULTI-Coord	[O]	Inter-unit coordination ledger construction

The full BIO endpoint E_{Bio} is now certified on the declared nondegenerate regime $\mathcal{C}^{\mathrm{nd}}_{\mathrm{Bio}}$. The scope restriction is the honest statement: this is replication-heredity closure on configurations where replication, metabolism, and external resource exchange are all genuinely present.

11. Developmental Differentiation via SCC Role Carriers

This section establishes the framework for developmental differentiation as stable role specialization in a coordinated BIO assembly, carried by minimal SCC-style support rather than imported cell types. No primitive "cell type," "developmental program," "organ," "tissue," or "fate map" is assumed. A role is lawful only when it is visible in the inter-unit ledgers.

11.1. Developmental Roles.

Definition 11.1 ([D]). (BIO Developmental Role.) Let $\mathcal{U} = \{U_1, \dots, U_k\}$ be a coordinated BIO assembly with inter-unit ledgers $\mathcal{M}^{\mathrm{inter}}_{\mathcal{U}}$, $\mathcal{H}^{\mathrm{inter}}_{\mathcal{U}}$, coordination graph $G_{\mathcal{U}}$, and second-order boundary $\mathcal{B}^{(2)}_{\mathrm{Bio}}$. The *developmental role* of a unit U_i is its quotient-visible contribution profile:

$$\rho(U_i) \;:=\; \big(\mathcal{M}^{\mathrm{inter}}_{\mathcal{U}}|_{U_i},\; \mathcal{H}^{\mathrm{inter}}_{\mathcal{U}}|_{U_i},\; \mathcal{B}^{(2)}_{\mathrm{Bio}}|_{U_i},\; G_{\mathcal{U}}|_{U_i}\big).$$

Two units have the same role iff their contribution profiles are equivalent under declared unit-level probes and transport bookkeeping. This is intentionally operational: a role is what a unit does in the certified assembly ledgers, not what a biologist labels it.

Definition 11.2 ([D]). (Role Differentiation.) A coordinated BIO assembly $\mathcal{U}$ exhibits *role differentiation* when there exist $U_i, U_j \in \mathcal{U}$ with $\rho(U_i) \not\sim \rho(U_j)$ under declared unit-level probes, and the distinction persists under declared inter-unit transport and hereditary descent.

Definition 11.3 ([D]). (SCC Role Carrier.) For a role ρ, its *SCC role carrier* K_ρ is the lawful faithful support carrier for the role observable $\tau_{\mathrm{BioRole}}(\rho)$, whose support consists of:

(1) role witness/outcome support;
(2) normalized role-type support;
(3) transported persistence and defect-history support;
(4) ORA/CTM/TIN/ledger compatibility support.

This is the SCC support-sufficiency template specialized to BIO roles. No primitive cell-type label may appear as load-bearing support.

Definition 11.4 ([D]). (Developmental Differentiation Trajectory.) A *developmental differentiation trajectory* is a finite sequence $\mathcal{U}_0 \to \mathcal{U}_1 \to \cdots \to \mathcal{U}_n$ of coordinated BIO assemblies such that each transition is recorded by declared inter-unit metabolic and hereditary ledgers, role profiles $\rho_t(U_i)$ are quotient-visible at each stage, SCC role carriers K_{ρ_t} persist or lawfully change under transport, and no primitive fate map or developmental program is imported.

11.2. Role Theorems.

Theorem 11.5 ([C]). (Role Visibility.) [Conditional on Thms. 9.5 and 9.6.] *If a unit's role is defined by its contribution profile $\rho(U_i)$, then role identity and role distinction are quotient-visible.*

Proof. Each component of $\rho(U_i)$ is already declared BIO endpoint data at first or second order: inter-unit metabolism, inter-unit heredity, second-order boundary contribution, and coordination graph incidence. The coordination graph is the finite record of certified metabolic and hereditary edges, not a primitive topology. Therefore role distinction is witnessed by declared probes over declared ledgers. If no declared probe separates two contribution profiles, no role distinction may be drawn. □ □

Theorem 11.6 ([C]). (No Primitive Cell-Type Import.) [Conditional on Thm. 11.5 and Sr. 3.1.] *No developmental role in a coordinated BIO assembly may depend on a primitive cell type, fate label, tissue label, or organismal program not visible in the contribution profile $\rho(U_i)$.*

Proof. Suppose two units agree on all role profile components but are assigned different developmental roles. The difference is not sourced by inter-unit metabolism, inter-unit heredity, boundary contribution, or coordination graph data. It is therefore a hidden target-side biological label. That violates the BIO no-hidden-channel discipline (O-BIO.NC, discharged) and the Book V branch-legitimacy rule excluding hidden supplementation. □ □

Theorem 11.7 ([C]). (SCC Role-Carrier Minimality.) [Conditional on SCC Support Sufficiency and SCC-MCS available for the BIO role observable $\tau_{\mathrm{BioRole}}(\rho)$ (`thm:scc-mcs`, SCC Branch).] *Every developmental role ρ has a unique minimal lawful SCC role carrier M_ρ up to role-carrier equivalence.*

Proof. Treat the role observable as an SCC-style structural predicate asking whether a unit's contribution profile realizes a stable role across assembly transport.

The required support is exactly the four SCC load-bearing components (Def. 11.3). By SCC Support Sufficiency, all lawful role support is exhausted by those four classes. By SCC Support-Rigidity and Shared-Support Faithfulness (SCC Branch), the common realized-support carrier remains faithful. SCC-MCS then gives uniqueness of the minimal carrier. This is essential for anti-plasticity: without minimality, a "role change" could be produced by adding hidden support; with M_ρ in hand, role transitions must be supported by the unique minimal carrier change. □ □

THEOREM 11.8 ([C]). (Differentiation Trajectory Legitimacy.) [Conditional on Thm. 11.7 and every role transition $\rho_t(U_i) \to \rho_{t+1}(U_i)$ being recorded by declared inter-unit ledgers.] *A developmental differentiation trajectory is lawful if every role transition is ledger-visible and carried by SCC-minimal role carriers.*

PROOF. Role transition is visible by Thm. 11.5. Hidden cell-type import is blocked by Thm. 11.6. Minimal role carriers are supplied by Thm. 11.7. Since each transition is ledger-visible and transport-accounted, the trajectory is source-descended and Book V-compliant.

The SCC contribution is essential: without MCS, a "role change" could be produced by adding hidden support outside the four declared classes. With M_ρ in hand, the role transition must be supported by the unique minimal carrier change, preventing arbitrary carrier expansion across stages. □ □

THEOREM 11.9 ([C]). (Conditional Developmental-Differentiation Endpoint.) [Conditional on Thms. 9.8, 11.5, 11.6, 11.7, 11.8; and the declared role-carrier-stable regime $\mathcal{C}^{\mathrm{dev}}_{\mathrm{Bio}}$ of coordinated BIO assemblies with quotient-visible role differentiation and SCC-minimal role carriers.] *The developmental endpoint datum*

$$E^{\mathrm{dev}}_{\mathrm{Bio}} \;=\; \left(E^{\mathrm{multi}}_{\mathrm{Bio}},\ \{\rho\},\ \{M_\rho\},\ \mathcal{T}_{\mathrm{dev}}\right)$$

is a lawful developmental-differentiation endpoint: developmental differentiation is certified as stable role specialization under certified coordination, not as imported developmental biology.

PROOF. The multicellular endpoint $E^{\mathrm{multi}}_{\mathrm{Bio}}$ supplies certified units, inter-unit coordination, and second-order boundary. Thm. 11.5 supplies role visibility. Thm. 11.6 blocks imported cell types. Thm. 11.7 supplies SCC-minimal anti-plasticity for roles. Thm. 11.8 supplies lawful role-transition trajectories. No organism-level primitive is added; everything is built from declared first-order units, second-order ledgers, and SCC-minimal role carriers. Book V branch legitimacy is satisfied: declared endpoint, observable family visibility, source descent, no-hidden supplement, and no path multiplication. □ □

REMARK 11.10 ([R]). **SCC architecture and anti-plasticity.** SCC's utility in the developmental setting is specific and structural: it provides the anti-plasticity mechanism that prevents role specialization from becoming arbitrary carrier inflation. Without SCC-minimal role carriers, one could always inflate a "role" by adding hidden support; the SCC branch's MCS theorem says the minimal faithful carrier is unique up to equivalence, blocking this inflation. The synthesis document is explicit on this point: SCC does not act as a consciousness or endpoint-label mechanism here; its sole structural contribution is anti-plasticity for role carriers.

Open Obligation 11.11 ([C]). (O-BIO.DEV-SCC: SCC Role-Carrier Dependency — Conditionally Resolved.) [Cascade from SCC-MCS discharge (Session XLII, `thm:scc-mcs`).] Thm. 11.7 requires SCC-MCS for BIO role observables. O-SCC.MCS has been conditionally discharged (`thm:scc-mcs`, SCC Branch) via the SCC-FL/LB/Zorn existence argument. The SCC-SV, SCC-TL, SCC-FL, SCC-LB chain is already certified [C] in the SCC branch. The branch-local verification required here is that BIO role observables satisfy SCC support sufficiency (four-component support: realized spectral/coercive, normalized type, transported defect-history, ORA/CTM/TIN/ledger compatibility).

By Def. 11.3, the BIO role carrier is defined with exactly those four support components. Support sufficiency (Thm. 11.5) is established: role identity and distinction are quotient-visible. Finite localizability follows because each role carrier component is finite at the certified BIO unit scope. Therefore the SCC-MCS conditions hold for BIO role observables, and O-BIO.DEV-SCC is conditionally discharged as a cascade from the SCC-MCS discharge.